

## 09 Batch Reactors

Batch reactor mole balance:

- In terms of number of moles ( $N$ ):  $\frac{dN_A}{dt} = r_A V$ . In words, the rate of change in the number of moles of species A changes according to the rate of generation of A times the system volume.

- If species A is a reactant, the rate of generation of A is *negative*

- If species A is a product, the rate of generation of A is *positive*

- Hence the balance equation "makes sense" because it properly accounts for the rate of change in the number of moles of the species of interest.

- In terms of conversion ( $X$ ):  $\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}}$

- Notice that this balance is written with respect to the rate of consumption of A. This makes sense because

*if A is a product, the rate of consumption of A ( $-r_A$ ) is positive, so  $X_A$  increases as time goes on.*

- Applying the batch reactor mole balances: If you are solving a reactor design problem for a prototypical batch reactor:

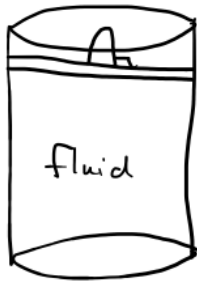
*you can write down the mole balance and just use it without re-deriving it.*

- In other words, we've already derived the mole balance for the prototypical batch reactor. You do not need to rederive it on your problem solutions, except for the following circumstances:

- The problem statement asks you to derive the balance. In this case, we are wanting to test your ability to derive a balance.
- The reactor is not a prototypical batch reactor. E.g., it could be a semibatch reactor. In that case, you should derive the appropriate balances for the relevant species, making sure to state your assumptions.

- There are (at least) two types of batch reactors: \_\_\_\_\_

Constant volume



rigid walls -  
if the number of moles changes during the reaction, the volume still stays constant

\*important for gas phase reactions

Constant pressure



non-rigid walls, e.g., a balloon

if the number of moles changes over the reaction the

- Let's do an example: The elementary gas phase reaction  $A + B \rightarrow C$  is carried out isothermally in a constant system volume batch reactor and in a constant pressure batch reactor. The initial mixture is equimolar in A and B, volume and no C is present initially in both cases. Which reactor takes longer to get to 60% conversion? changes

- First: Practice with the concept! Pause the video, and try answering conceptually. Why did you choose to the answer you did? How confident are you in your response?

Constant volume reactor takes longer!

maintain constant P.

in the constant P reactor:  $\downarrow N_T \downarrow V \uparrow C \uparrow \text{rate}$

- How would this answer change if this were a liquid phase reaction? in liquid phase, we assume constant volume, so both cases would be the same.
- Now, let's do the numerical solution:

① constant volume  $\rightarrow V = V_0$

$$\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}} \quad \cdot \quad -r_A = k C_A C_B \cdot$$

$$C_A = C_{A0} (1 - X_A) = C_B$$

$$-r_A = k C_{A0}^2 (1 - X_A)^2$$

$$\frac{dX_A}{dt} = \frac{k C_{A0}^2 (1 - X_A)^2 V}{N_{A0}}, \quad \text{where } C_{A0} = \frac{N_{A0}}{V_0} = \frac{N_{A0}}{V}$$

$$\frac{dX_A}{dt} = k C_{A0} (1 - X_A)^2$$

$$\int_0^{0.6} \frac{dX_A}{(1 - X_A)^2} = k C_{A0} \int_0^t dt$$

$t = \frac{1.5}{k C_{A0}}$

constant volume  
solution

Constant pressure

$$V_0 = \frac{N_{T0} R T_0}{P_0}$$

$$V = \frac{N_T R T}{P}$$

$$T = T_0, \quad P = P_0$$

book definition

$$V = V_0 \left( \frac{N_T}{N_{T0}} \right) \rightarrow \frac{N_T}{N_{T0}} = 1 + \sum X$$

$\nwarrow y_{A0} \delta$

$$Y_{A0} = \frac{1}{2}, \quad \xi = \left(\frac{1}{2}\right)(-1)$$

$$\frac{N_T}{N_{T0}} = 1 - \frac{X_A}{2}$$

- rate law is the same as in the constant  $V$  case

$$-r_A = k C_A C_B$$

$$C_A = \frac{C_{A0}(1-X_A)}{\left(1-\frac{X_A}{2}\right)} = C_B$$

$$\frac{dX_A}{dt} = \frac{-r_A V}{N_{A0}} = \frac{k C_{A0}^2 (1-X_A)^2 V}{\left(1-\frac{X_A}{2}\right)^2 N_{A0}}$$

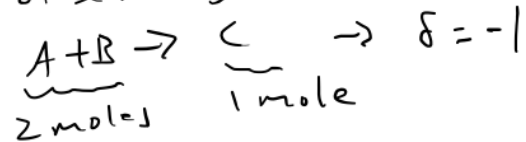
$$\frac{dX_A}{dt} = \frac{k C_{A0}^2 (1-X_A)^2 V_0}{\left(1-\frac{X_A}{2}\right) N_{A0}} = \frac{k C_{A0} (1-X_A)^2}{\left(1-\frac{X_A}{2}\right)}$$

$$\int_0^t \frac{\left(1-\frac{X_A}{2}\right)}{(1-X_A)^2} dX_A = k C_{A0} \int_0^t dt$$

$$t = \frac{1.208}{k C_{A0}}$$

constant pressure  
case

↑  
mole fraction of the limiting reactant at  $t=0$  is the change in the number of moles due to the reaction, per mole of limiting reactant



$$V = V_0 \left(1 - \frac{X_A}{2}\right)$$